ELE214 - Electronics Laboratory I Preliminary Work 5

1 Questions

1. Explain how you can determine the type of a MOSFET (n– or p–channel) and test whether it is defective or not.

2. Sketch $I_D - V_{DS}$ graph for $V_{GS} = 1.2, 1.4, 1.6, \ldots, 2.2$ V.

3. Sketch $I_D - V_{GS}$ graph, indicate V_{TH} using output characteristics.

Experimental Work (Simulation of 1-3)

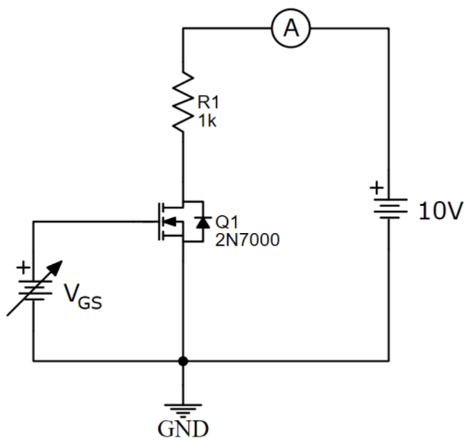


Figure 1: Circuit diagram for output characteristics

1. Test the 2N7000 MOSFET and determine its type.

2. Considering *Figure 1*, take the V_{GS} constant to some value. Vary V_{DD} in steps of 1 V up to 10 V and measure the I_D. Repeat for $V_{GS} = 1.4, 1.6, \ldots, 2.2$ V. Obtain the $I_D - V_{DS}$ characteristics.

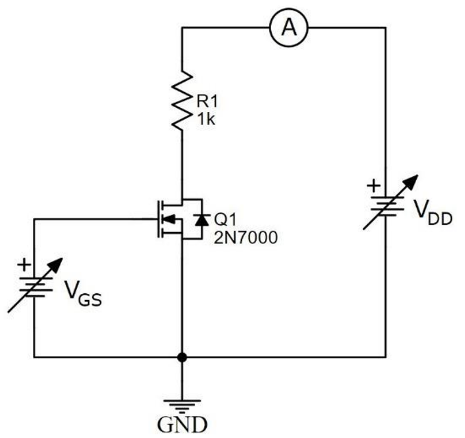


Figure 2: Circuit diagram for transfer characteristics

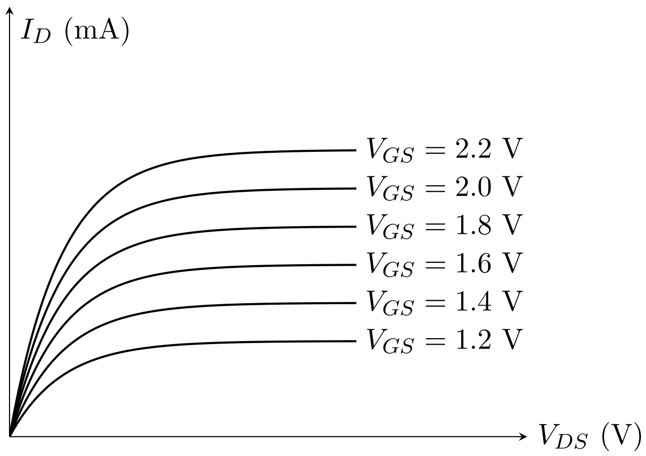
3. Considering *Figure 2*, obtain the $I_D - V_{GS}$ characteristics. Vary V_{GS} in the step of 0.2 V up to 2.4 V and measure the I_D. Set the $V_{DD} = 10$ V. Observe the critical point (V_{TH}).

2 Answers

1. The type of a MOSFET can be identified by using a digital multimeter in diode test mode to check the internal body diode between the drain and source terminals. In an n-channel MOSFET, the body diode conducts when the red probe is connected to the source and the black probe to the drain, while in a p-channel MOSFET conduction occurs when the red probe is connected to the drain and the black probe to the source.

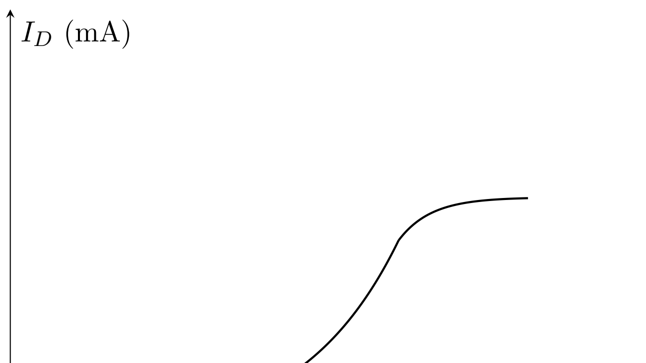
To test whether the MOSFET is defective, the resistance between the terminals is checked. The gate should be insulated from both the drain and source, showing very high resistance. A low resistance between gate and source or between drain and source usually indicates a faulty MOSFET. The switching action can also be tested by charging the gate and observing whether the drain-source resistance changes from high (OFF state) to low (ON state). If the MOSFET fails to switch properly or remains permanently shorted or open, it is defective.

2. The graph is given below.



Graph 2: Transfer characteristics for the given voltages

3. The expected graph is given below.



Graph 3: Output characteristics for the given voltages

Simulation

Since the model isn't included in LTspice internally, the directive `.model 2N7000 VDMOS(Rg=3 Vto=1.6 Rd=0 Rs=.75 Rb=.14 Kp=.17 mtriode=1.25 Cgdmax=80p Cgdmin=12p Cgs=50p Cjo=50p Is=.04p mfg=Fairchild Vds=60 Ron=2 Qg=1.5n)` is used.

1. An AC source is applied between the source and drain terminals.

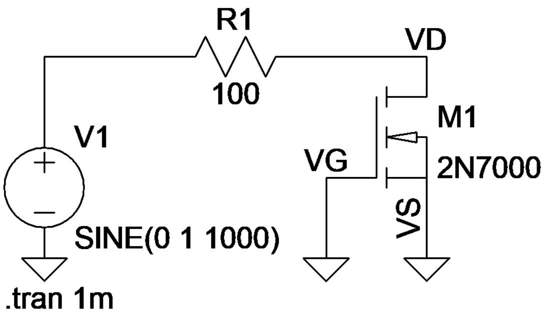


Figure 3: MOSFET Defect Test Circuit

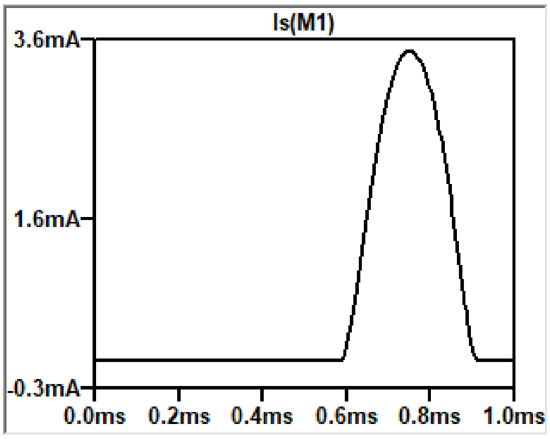


Figure 4: Source Current in the Test

As expected, the MOSFET is working properly.

2. Since the source is grounded, we can measure the drain voltage. directly.

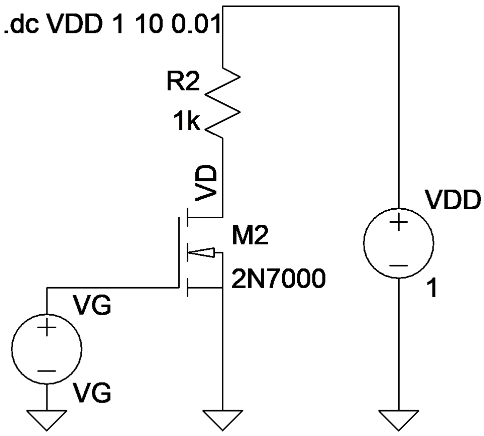


Figure 5: Adjusted Circuit

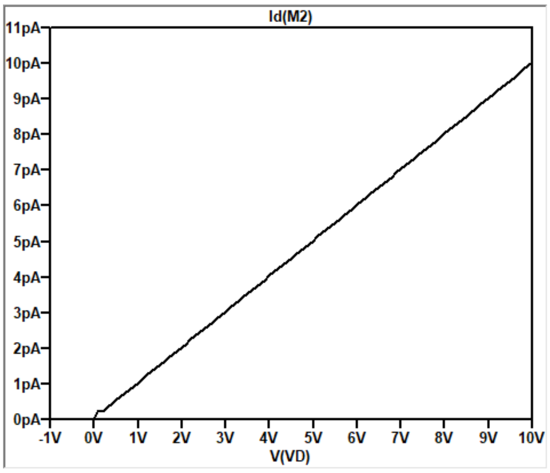


Figure 6: Drain Current for $V_{GS} = 1.4$ V

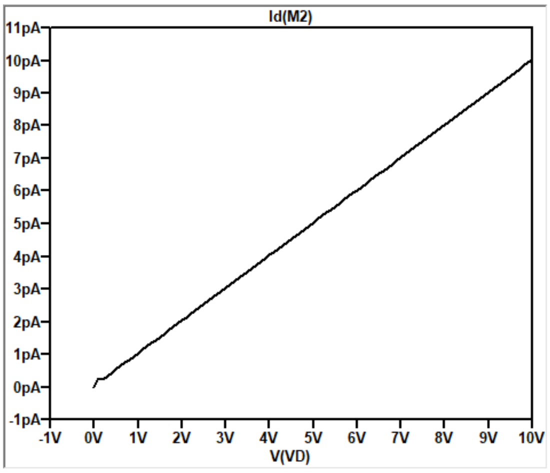


Figure 7: Drain Current for $V_{GS} = 1.6$ V

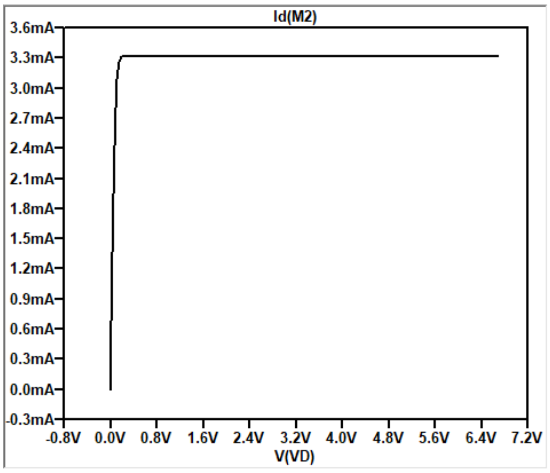


Figure 8: Drain Current for $V_{GS} = 1.8$ V

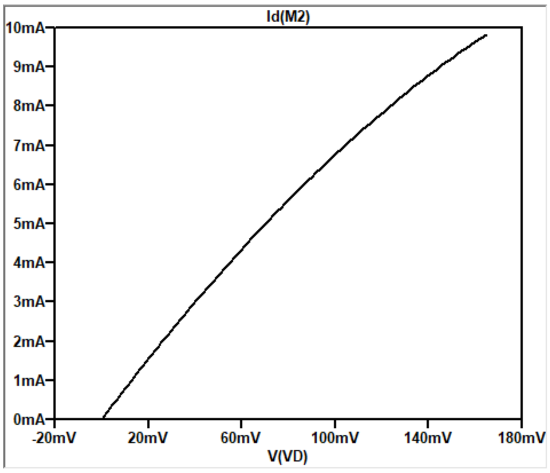


Figure 9: Drain Current for $V_{GS} = 2.0$ V

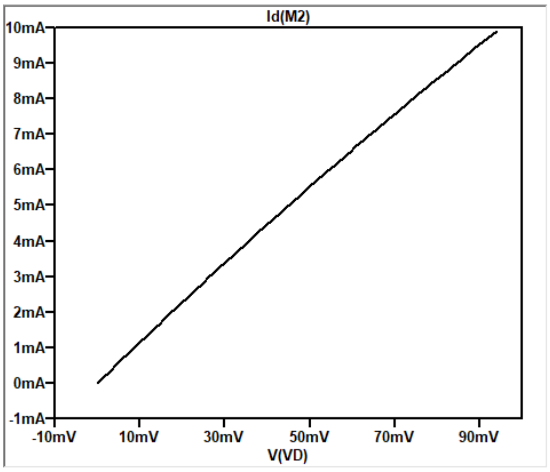


Figure 10: Drain Current for $V_{GS} = 2.2$ V

Table 1: V_{DS} (V) and I_D (mA) values

$V_{GS} = 1.4$ V		$V_{GS} = 1.6$ V		$V_{GS} = 1.8$ V		$V_{GS} = 2$ V		$V_{GS} = 2.2$ V	
V_{DS}	I_D	V_{DS}	I_D	V_{DS}	I_D	V_{DS}	I_D	V_{DS}	I_D
0	0	0	0	0	0	0	0	0	0
1	$1.0 \cdot 10^{-9}$	1	$1.0 \cdot 10^{-9}$	1	0.97	1	0.99	1	0.99
2	$2.0 \cdot 10^{-9}$	2	$2.0 \cdot 10^{-9}$	2	1.94	2	1.97	2	1.98
3	$3.0 \cdot 10^{-9}$	3	$3.0 \cdot 10^{-9}$	3	2.89	3	2.96	3	2.97
4	$4.0 \cdot 10^{-9}$	4	$4.0 \cdot 10^{-9}$	4	3.32	4	3.95	4	3.96
5	$5.0 \cdot 10^{-9}$	5	$5.0 \cdot 10^{-9}$	5	3.32	5	4.93	5	4.96
6	$6.0 \cdot 10^{-9}$	6	$6.0 \cdot 10^{-9}$	6	3.32	6	5.91	6	5.95
7	$7.0 \cdot 10^{-9}$	7	$7.0 \cdot 10^{-9}$	7	3.32	7	6.90	7	6.94
8	$8.0 \cdot 10^{-9}$	8	$8.0 \cdot 10^{-9}$	8	3.32	8	7.88	8	7.93
9	$9.0 \cdot 10^{-9}$	9	$9.0 \cdot 10^{-9}$	9	3.32	9	8.86	9	8.92
10	$10.0 \cdot 10^{-9}$	10	$10.0 \cdot 10^{-9}$	10	3.32	10	9.61	10	9.91

3. $V_{TH} = 1.6$ V. The circuit is given below.

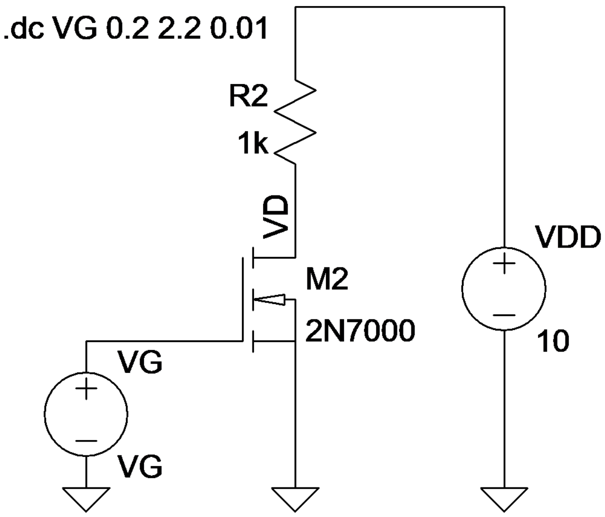


Figure 11: Adjusted Circuit

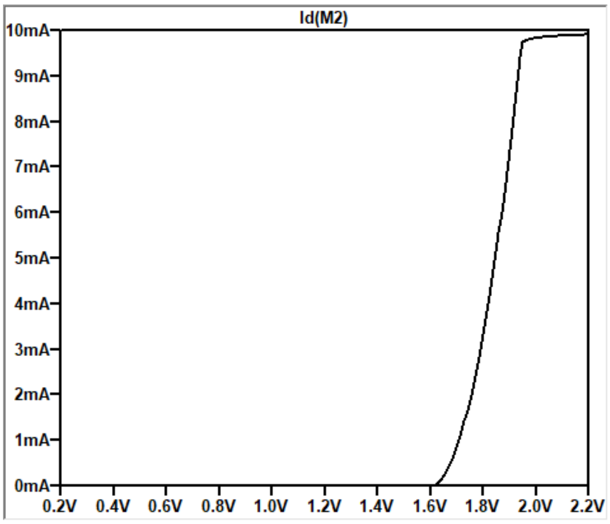


Figure 12: I_D–V_{GS} characteristics

Table 2: I_D Values for Different Values of V_{GS}

	V_{GS} (V)	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2
$V_{DD} = 10$ V	I_D (mA)	10^{-8}	10^{-8}	10^{-8}	10^{-8}	10^{-8}	10^{-8}	10^{-8}	10^{-8}	3.32	9.84	9.91